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(54) **PROBE UNIT AND CONTACT PROBE**

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ABSTRACT

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A probe unit includes: a plurality of probe groups each including two contact probes configured to come into contact with one electrode of a contact target; and a probe holder configured to hold the contact probes. Each of the two contact probes includes: a distal end portion; and a flange portion extending from a proximal end side of the distal end portion and having a maximum length in a direction orthogonal to a longitudinal axis of the contact probe larger than a maximum length of the distal end portion, the flange portion including a first flat surface portion having a planar shape provided on a part of a side surface of the flange portion, and the holder hole has a stepped shape configured to be locked by the flange portion and has a wall surface abutting on the first flat surface portion.

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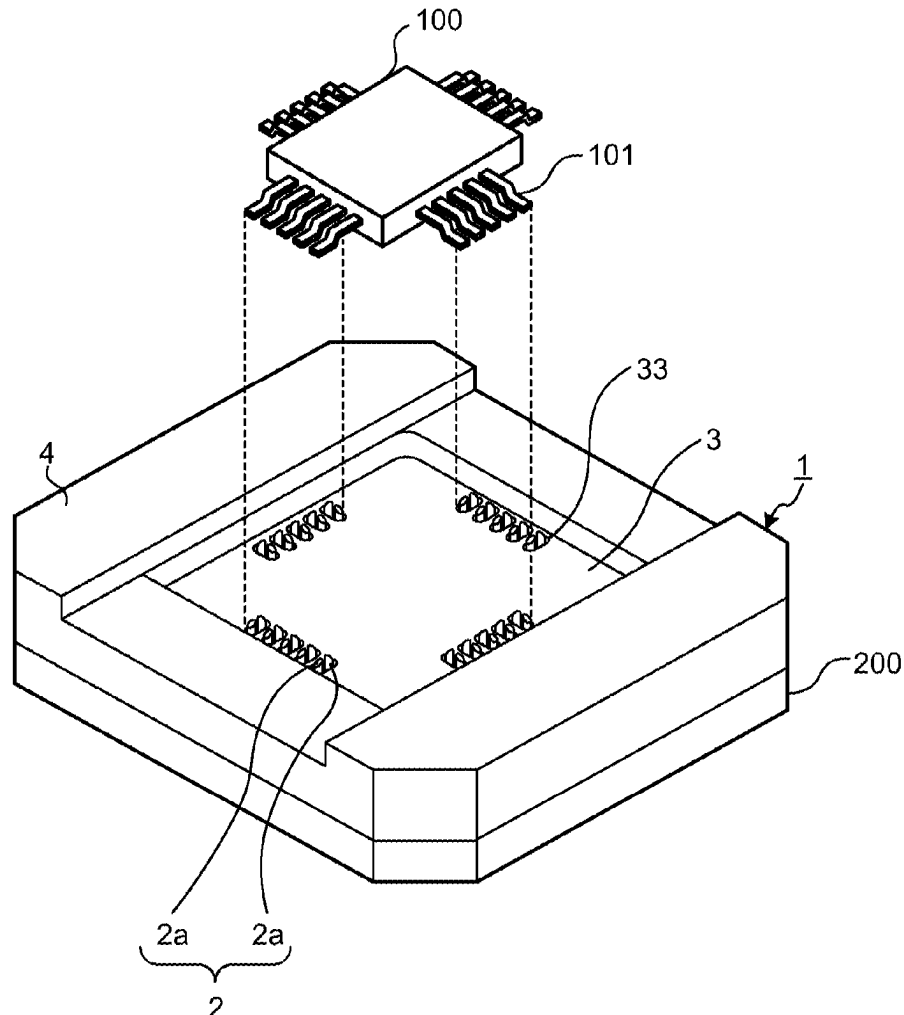


FIG.1

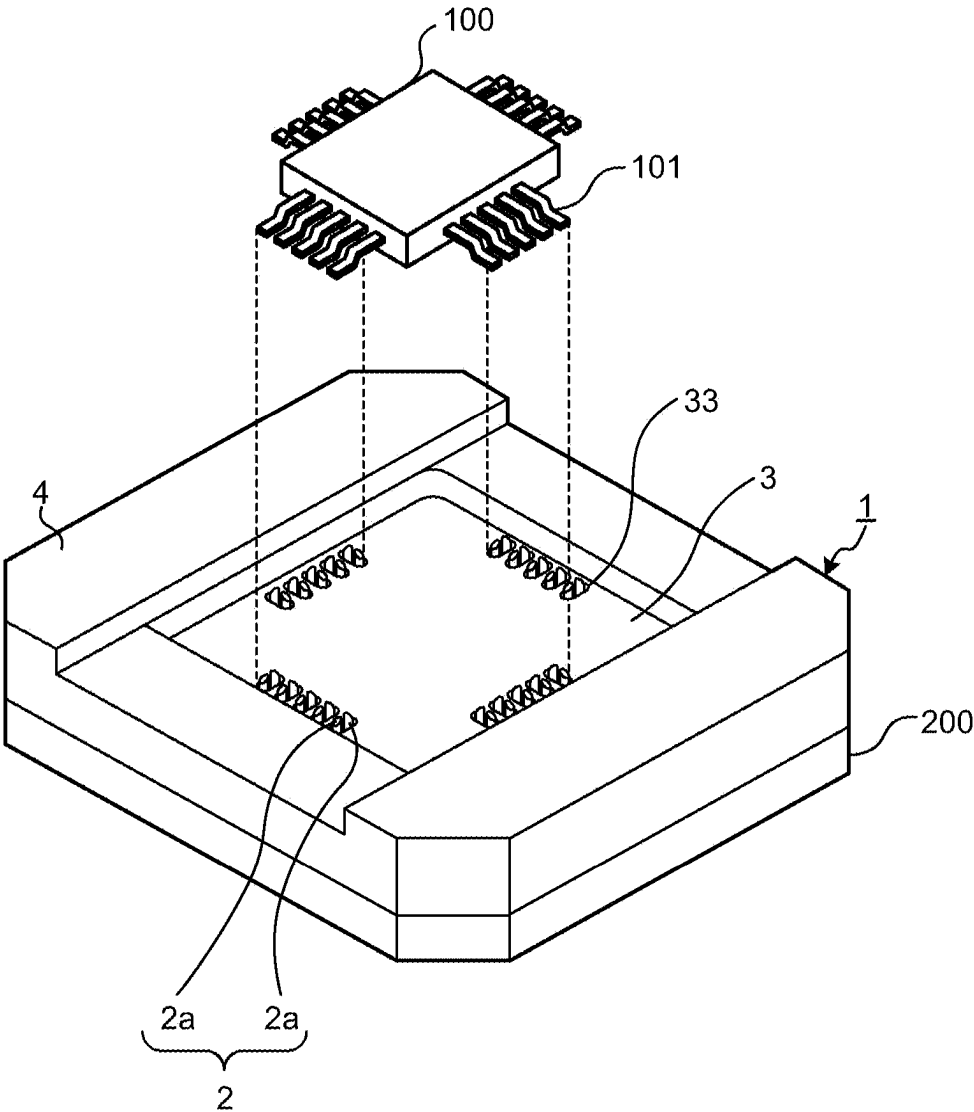


FIG.2

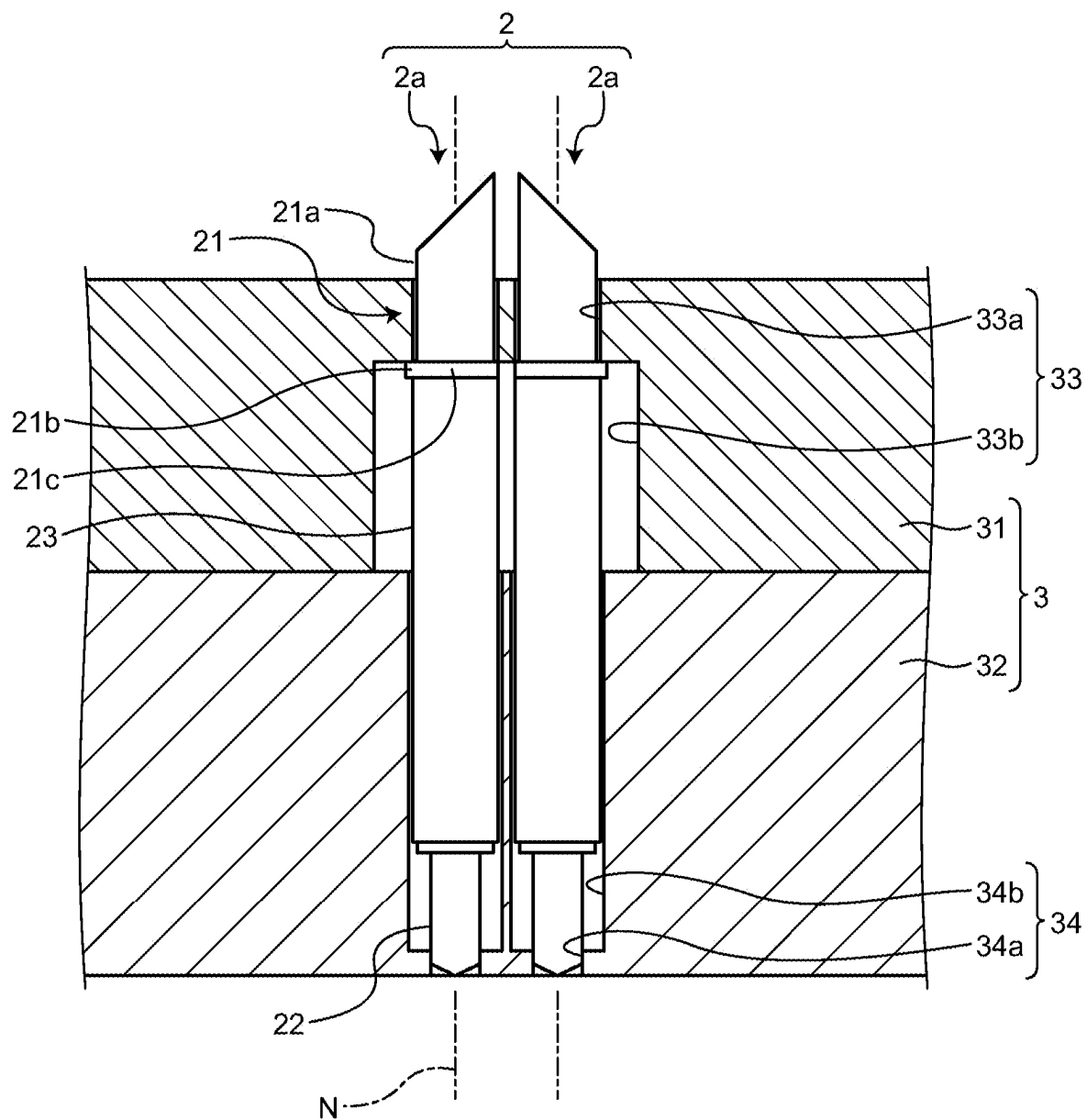


FIG.3

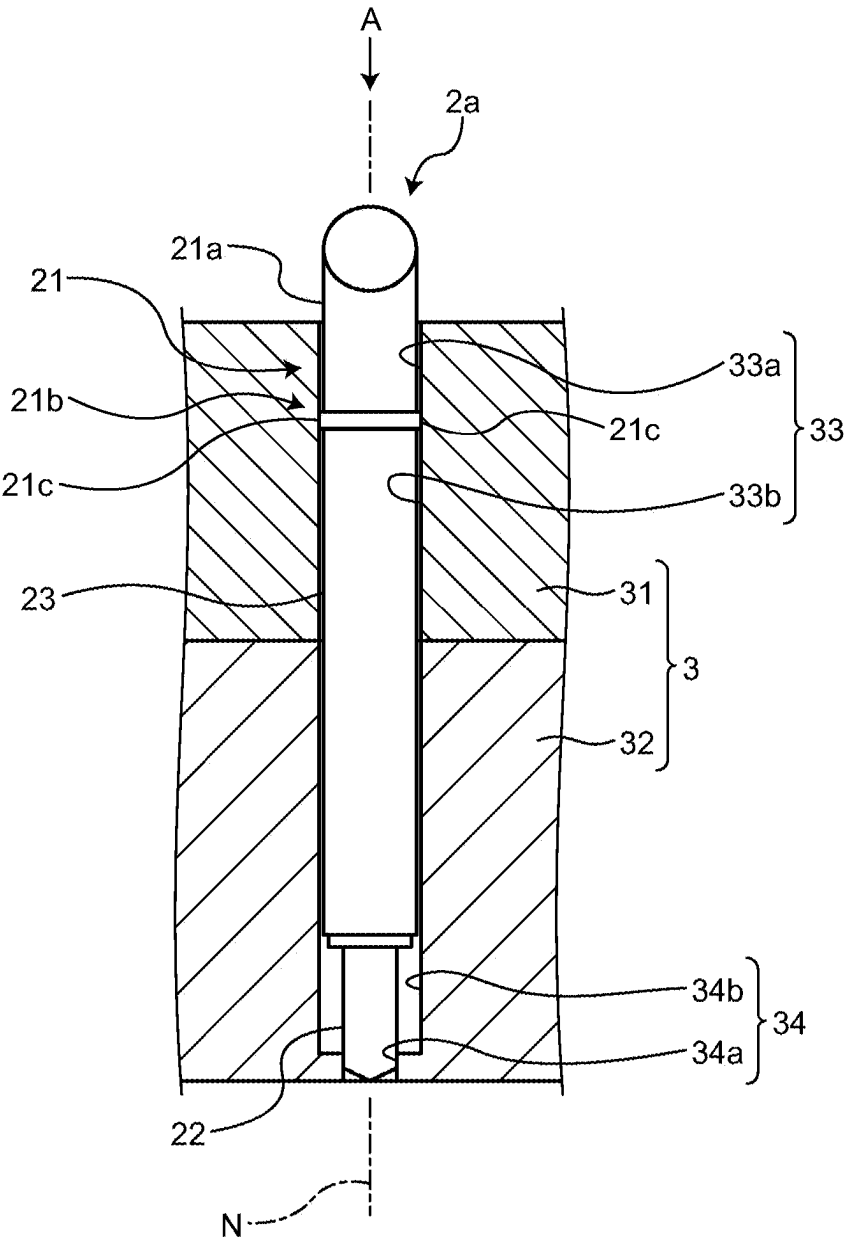


FIG.4

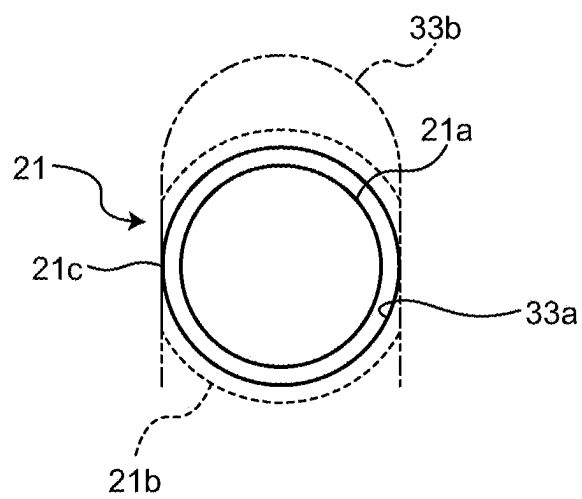


FIG.5

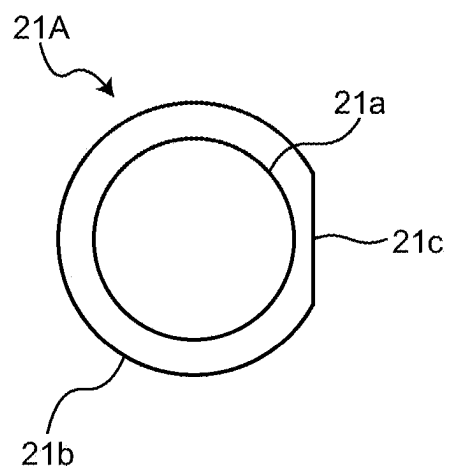


FIG.6

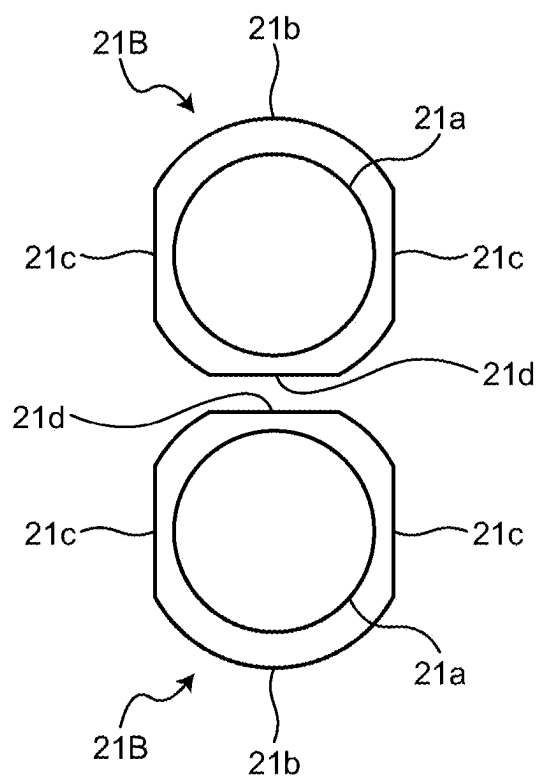


FIG.7

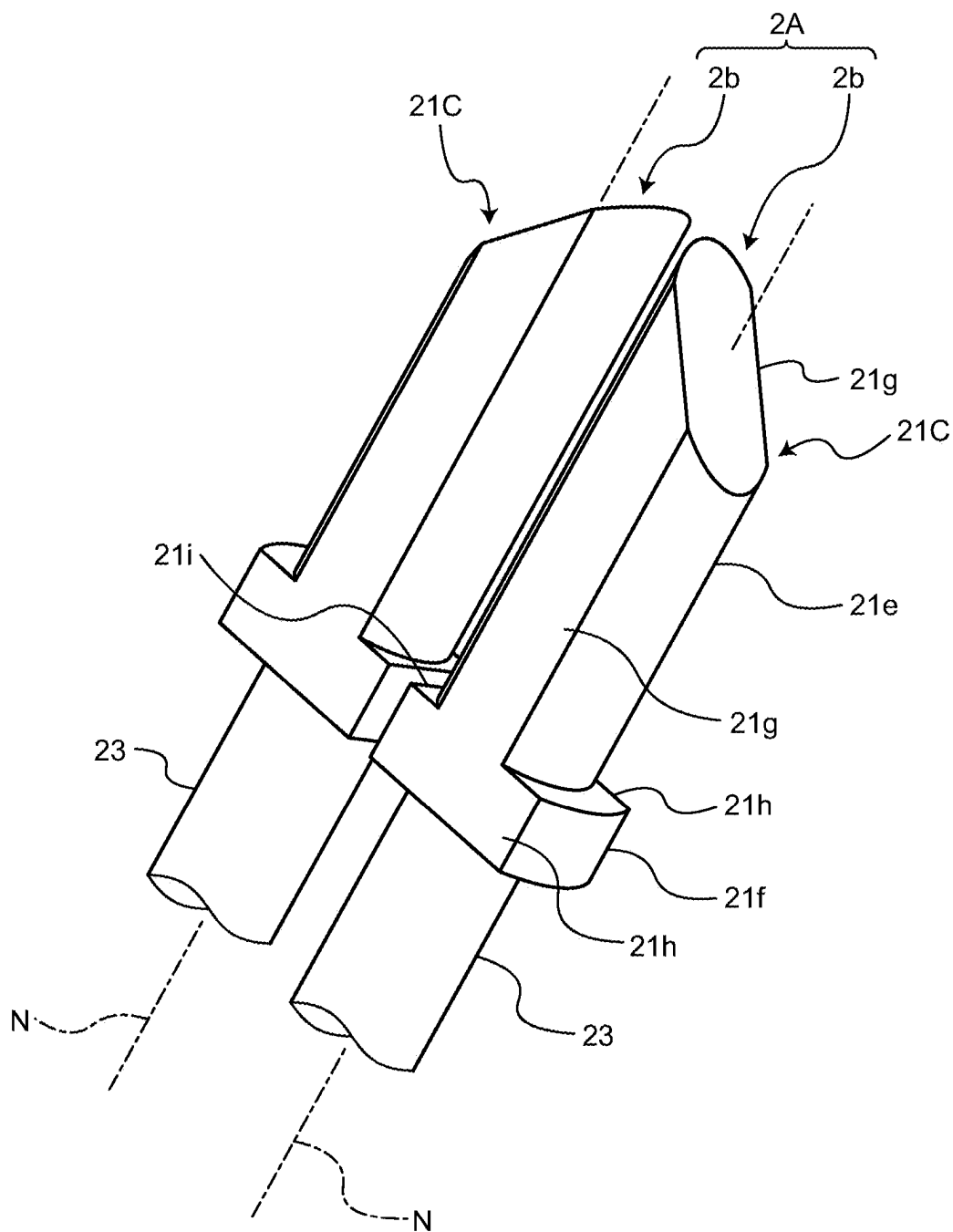
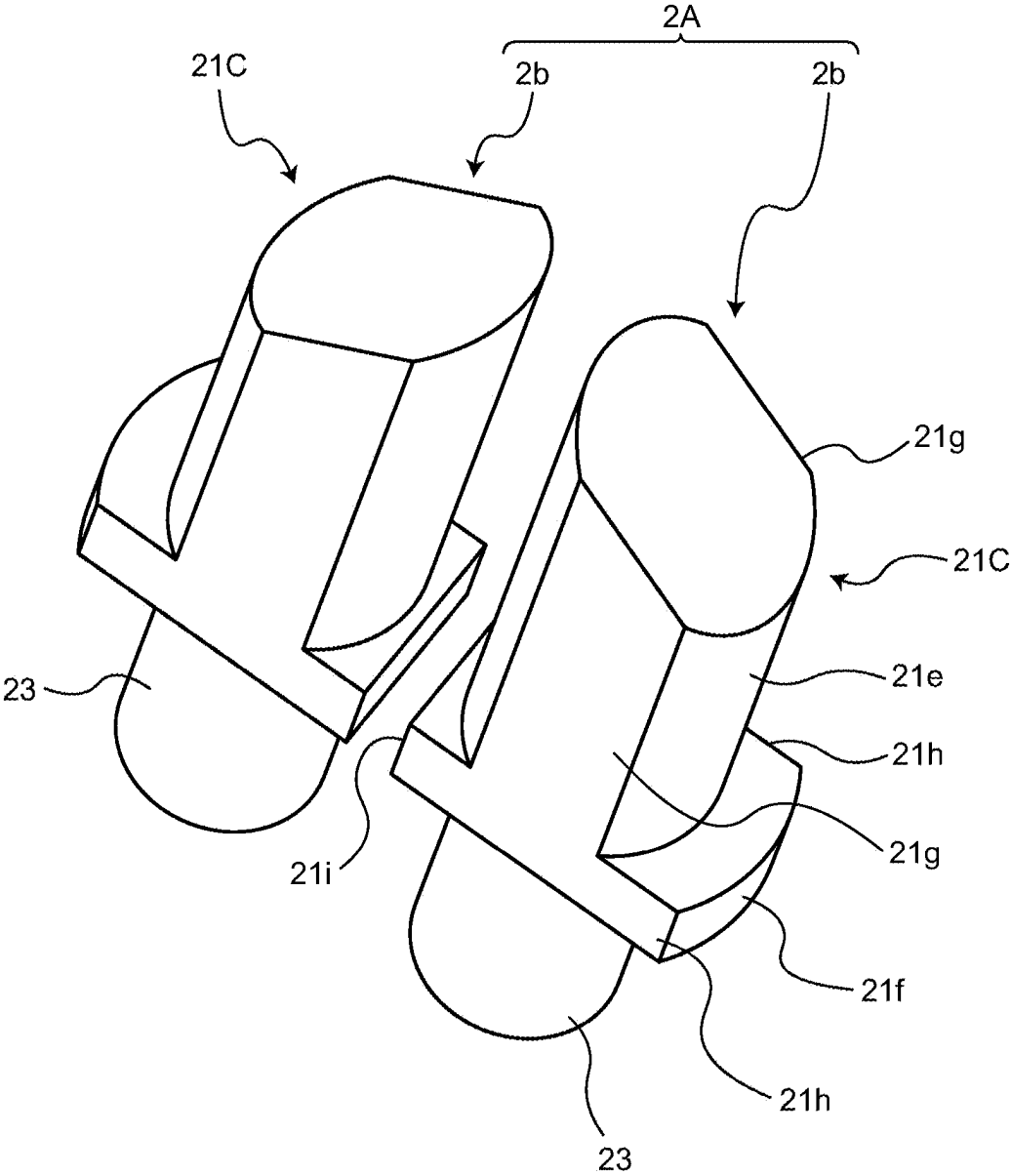


FIG.8



PROBE UNIT AND CONTACT PROBE

[0001] This application is based upon and claims the benefit of priority of the prior Japanese Patent Application No. 2024-035755, filed on Mar. 8, 2024, the entire contents of which are incorporated herein by reference.

BACKGROUND

[0002] The present disclosure relates to a probe unit and a contact probe used for conduction state inspection or operation characteristic inspection of an inspection target such as a semiconductor integrated circuit or a liquid crystal panel.

[0003] In the related art, when conduction state inspection or operation characteristic inspection of an inspection object such as a semiconductor integrated circuit or a liquid crystal display device is performed, a conductive contact probe that electrically connects the inspection target and a signal processing device having a circuit board that outputs an inspection signal is used. As one of the above-described conduction state inspection and operation characteristic inspection methods, there is a four-terminal measurement method. As a probe unit using this four-terminal measurement method, a technique is disclosed in which a distal end of each contact probe is brought into contact with an inspection target to measure an electrical characteristic by a set of contact probes (probe group) held by a probe holder (see, for example, JP 2021-105550 A).

SUMMARY

[0004] In the technique disclosed in JP 2021-105550 A, in order to prevent the contact probe from rotating with respect to the probe holder, the distal end portion is formed in a flat plate shape, and the flat surface portion is brought into contact with the holder. For this reason, the volume of the distal end portion is smaller than that of other portions, and there is a problem that durability is deteriorated due to repetition of inspection or the like.

[0005] There is a need for a probe unit and a contact probe capable of securing strength of the contact probe while suppressing rotation with respect to a probe holder.

[0006] According to one aspect of the present disclosure, there is provided a probe unit including: a plurality of probe groups each including two contact probes configured to come into contact with one electrode of a contact target on one end portion side in a longitudinal direction, each of the two contact probes coming into contact with a different electrode of a substrate on another end portion side; and a probe holder configured to hold the contact probes, the probe holder including a plurality of holder holes configured to hold the plurality of contact probes, wherein each of the two contact probes includes: a distal end portion configured to come into contact with the one electrode of the contact target at a distal end; and a flange portion extending from a proximal end side of the distal end portion and having a maximum length in a direction orthogonal to a longitudinal axis of the contact probe larger than a maximum length of the distal end portion, the flange portion including a first flat surface portion having a planar shape provided on a part of a side surface of the flange portion, and the holder hole has a stepped shape configured to be locked by the flange portion and has a wall surface abutting on the first flat surface portion.

[0007] According to another aspect of the present disclosure, there is provided a contact probe for coming into

contact with one electrode of a contact target on one end portion side in a longitudinal direction, the contact probe including: a distal end portion configured to come into contact with the one electrode of the contact target at a distal end; and a flange portion extending from a proximal end side of the distal end portion and having a maximum length in a direction orthogonal to a longitudinal axis of the contact probe larger than a maximum length of the distal end portion, the flange portion including a flat surface portion having a planar shape provided on a part of a side surface of the flange portion.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a perspective view illustrating a configuration of a probe unit according to an embodiment;

[0009] FIG. 2 is a partial cross-sectional view illustrating a configuration of a main portion of a probe unit according to the embodiment;

[0010] FIG. 3 is a partial cross-sectional view illustrating a configuration of a main portion of a probe unit according to the embodiment;

[0011] FIG. 4 is a view illustrating a configuration of a first plunger and a probe holder viewed from a direction of an arrow A in FIG. 3;

[0012] FIG. 5 is a view for explaining a configuration of a first plunger according to a first modification;

[0013] FIG. 6 is a view for explaining a configuration of a first plunger according to a second modification;

[0014] FIG. 7 is a perspective view (part 1) for explaining a configuration of a probe according to a third modification; and

[0015] FIG. 8 is a perspective view (part 2) for explaining a configuration of the probe according to the third modification.

DETAILED DESCRIPTION

[0016] Hereinafter, embodiments will be described in detail with reference to the drawings. Note that the present disclosure is not limited by the following embodiments. In addition, the drawings referred to in the following description merely schematically illustrate shapes, sizes, and positional relationships to the extent that the contents of the present disclosure may be understood. That is, the present disclosure is not limited only to the shape, size, and positional relationship illustrated in each drawing.

[0017] FIG. 1 is a perspective view illustrating a configuration of a probe unit according to an embodiment. A probe unit 1 illustrated in FIG. 1 is a device used when an electrical characteristic inspection of a semiconductor integrated circuit as an inspection object is performed, and is a device that electrically connects a connection electrode of a semiconductor package 100 enclosing the semiconductor integrated circuit and a circuit board 200 that outputs an inspection signal to the semiconductor integrated circuit. The connection electrode of the semiconductor package 100 is a lead 101 illustrated in FIG. 1 and is connected to a semiconductor integrated circuit.

[0018] A probe unit 1 includes: a probe holder 3 that includes a plurality of probe groups 2 including two contact probes 2a (hereinafter, simply referred to as "probe 2a") that each come into contact with one connection electrode (lead 101) of a semiconductor package 100 that is a contacted body on one end portion side in a longitudinal direction and

comes into contact with different electrodes of a circuit board **200** on the other end portion side, and accommodates and holds the plurality of probe groups **2** according to a predetermined pattern; and a holder member **4** that is provided around the probe holder **3** and suppresses occurrence of positional deviation of the semiconductor package **100** that comes into contact with the plurality of probe groups **2** at the time of inspection.

[0019] FIG. 2 is a partial cross-sectional view illustrating a configuration of a main portion of a probe unit according to an embodiment. FIG. 3 is a partial cross-sectional view illustrating a configuration of a main portion of a probe unit according to an embodiment. FIG. 4 is a view illustrating a configuration of a first plunger and a probe holder viewed from a direction of an arrow A in FIG. 3. FIG. 2 is a partial cross-sectional view illustrating a detailed configuration of the probe group **2** accommodated in the probe holder **3**, and is a cross sectional view taken along a flat surface parallel to an arrangement direction of the probe group **2**. FIG. 3 is a partial cross-sectional view illustrating a detailed configuration of the probe group **2** accommodated in the probe holder **3**, and is a cross sectional view taken along a flat surface orthogonal to an arrangement direction of the probe group **2**. In the probe group **2** illustrated in FIGS. 2 and 3, two probes **2a** formed using a conductive material are arranged in parallel so as to have the same height. In addition, the probe groups **2** are arranged according to the arrangement of the connection electrodes (leads **101**) of the semiconductor package **100**.

[0020] The probe **2a** includes a first plunger **21** that comes into contact with the lead **101** of the semiconductor package **100** when the semiconductor integrated circuit is inspected, a second plunger **22** that comes into contact with the electrode of the circuit board **200** including the inspection circuit, and a barrel **23** to which the first plunger **21** is attached at one end and to which the second plunger **22** is extendably coupled at the other end. The first plunger **21** and the second plunger **22** constituting the probe **2a**, and the barrel **23** have the same axis (here, the axis N). The axis N corresponds to the longitudinal axis of the probe **2a**.

[0021] The first plunger **21** has a columnar distal end portion **21a** in which a flat surface inclined with respect to the axis is formed, and a flange portion **21b** extending from the proximal end side of the distal end portion **21a**, at least a part of which has a larger length in a direction orthogonal to the axis than the distal end portion **21a**, substantially coaxially. The first plunger **21** has a stepped shape at a connection portion between the distal end portion **21a** and the flange portion **21b**. Note that the first plunger **21** is attached to the barrel **23** by fitting, welding, or the like.

[0022] Note that “the same axis” and “coaxially” include manufacturing errors or the like.

[0023] The distal end portion **21a** has an inclined surface obliquely cut with respect to the axis N direction, and is in contact with the lead **101** of the semiconductor package **100** at the distal end.

[0024] The distal end portion **21a** has a cylindrical shape except for the distal end portion. That is, in the distal end portion **21a**, a portion other than the contact portion with the lead **101** forms a columnar shape without a notch, and the volume of the distal end portion **21a** is secured.

[0025] Note that the distal end portion **21a** may have any shape that may come into contact with the lead **101**, such as a crown shape or a cylindrical shape.

[0026] The flange portion **21b** has two flat surface portions **21c** formed by cutting a part of a side surface into a planar shape. The two flat surface portions **21c** are provided on opposite sides to each other with respect to the axis of the flange portion **21b**. Note that the axis of the flange portion **21b** is an axis extending in parallel with the axis of the first plunger **21** and passing through the center of gravity, and coincides with the axis of the first plunger **21** here.

[0027] In the present embodiment, the flat surface portion **21c** is provided on the side portion side of the inclined surface of the distal end portion **21a**.

[0028] Note that the flat surface portion **21c** may have an uneven shape at a plurality of locations in the circumferential direction to such an extent that the flat surface portion **21c** may come into contact with the wall surface of the probe holder **3**. That is, in the present embodiment, the “flat surface” of the flat surface portion **21c** includes a flat surface having surface roughness that allows contact with the wall surface of the probe holder **3** at a plurality of positions.

[0029] The second plunger **22** has a columnar shape having a tapered distal end shape. The second plunger **22** is in contact with a coil spring in the barrel **23** or is provided with a flange or the like to be prevented from coming off from the barrel **23**.

[0030] A coil spring is provided inside the barrel **23** to bias the second plunger **22**. When the semiconductor package **100** is brought into contact with the probe **2a**, the second plunger **22** moves in the axis direction with respect to the barrel **23** to relieve an impact on the connection electrode of the semiconductor package **100**, and applies a load to the semiconductor package **100** and the circuit board **200** by the biasing force of the coil spring in the barrel **23**.

[0031] The probe holder **3** is formed using an insulating material such as resin, machinable ceramic, or silicon, and is formed by laminating a first member **31** located on the upper surface side and a second member **32** located on the lower surface side in FIG. 2. A holder hole including a first hole portion **33** and a second hole portion **34** for accommodating the plurality of probes **2a** is formed in the first member **31** and the second member **32**. The formation position of the holder hole is determined according to the wiring pattern of the semiconductor package **100**. At this time, for example, the probe group **2** is arranged such that the distal end sides of the distal end portions **21a** of the probes **2a** approach each other.

[0032] Both the first hole portion **33** and the second hole portion **34** have stepped hole shapes having different diameters along the penetration direction. That is, the first hole portion **33** has an opening in the upper end surface of the probe holder **3**, and includes two small diameter portions **33a** through which the distal end portions **21a** of the probes **2a** passes, and a large diameter portion **33b** having a diameter larger than those of the small diameter portions **33a** and through which the two probes **2a** pass. The small diameter portion **33a** is slightly larger than the diameter of the distal end portion **21a** and smaller than the maximum length of the flange portion **21b**. In addition, the large diameter portion **33b** has a long hole shape capable of accommodating the two probes **2a**, and the shortest length is equal to or slightly larger than the distance between the flat surface portions **21c** of the flange portion **21b**.

[0033] On the other hand, the second hole portion **34** includes a small diameter portion **34a** having an opening in the lower end surface of the probe holder **3** and a large

diameter portion **34b** having a diameter larger than that of the small diameter portion **34a**. The small diameter portion **34a** has a diameter slightly larger than the diameter of the second plunger **22**. In addition, the large diameter portion **34b** has a diameter slightly larger than the diameter of the barrel **23**. The second hole portion **34** accommodates the probe **2a**. For this reason, in the probe group **2**, the probes **2a** are respectively accommodated in the through-holes in which the two second hole portions **34** are connected to the large diameter portion **33b** with respect to the first hole portion **33** having the two small diameter portions **33a** and the large diameter portion **33b**. The shapes of the first hole portion **33** and the second hole portion **34** are determined according to the configuration of the probe **2a** to be accommodated. Note that the large diameter portion **34b** through which the probes **2a** forming a set are inserted may be a common hole portion having no wall surface.

[0034] The flange portion **21b** of the first plunger **21** abuts on a boundary wall surface between the small diameter portion **33a** and the large diameter portion **33b** of the first hole portion **33**, thereby having a function of preventing the probe **2a** from coming out of the probe holder **3** (see FIG. 2). At this time, the flat surface portion **21c** abuts on a wall surface (here, a flat surface part of the long hole shape) of the large diameter portion **33b** or faces the wall surface with a slight gap (see FIG. 4). For this reason, when a force rotating around the axis **N** is applied to the probe **2a**, the flat surface portion **21c** abuts on the wall surface of the large diameter portion **33b**, whereby the rotation around the axis **N** is suppressed.

[0035] During the inspection of the semiconductor integrated circuit, the inspection signal supplied from the circuit board **200** to the semiconductor integrated circuit reaches the lead **101** of the semiconductor package **100** from the electrode of the circuit board **200** via the probes **2a**. Specifically, the probe **2a** reaches the lead **101** of the semiconductor package **100** via the second plunger **22**, the barrel **23**, and the first plunger **21**. Note that, in the two electrodes of the circuit board **200**, for example, one electrode is a measurement electrode (Sense), and the other electrode is a power transmission electrode (Force).

[0036] In addition, since the distal end of the distal end portion **21a** is formed to be tapered, even in a case where an oxide film is formed on the surface of the lead **101**, the oxide film may be broken through, and the distal end of the distal end portion **21a** may be brought into direct contact with the lead **101**.

[0037] According to the above-described embodiment, in the flange portion **21b** of the columnar first plunger **21**, two planar flat surface portions **21c** are formed, and the flat surface portions **21c** abut on the wall surface of the hole of the probe holder **3** (here, the wall surface of the large diameter portion **33b**). Therefore, by controlling the rotation of the probe **2a** around the axis while securing the volume of the distal end portion **21a**, it is possible to secure the strength of the contact probe while suppressing the rotation with respect to the probe holder.

[0038] In addition, according to the present embodiment, since the small diameter portion **33a** has a round hole shape and regulates the position of the distal end portion **21a**, a change in the distance between the distal end portions **21a** of the probes **2a** forming a pair is suppressed, and as a result, the positionability of the probe **2a** may be improved.

[0039] Next, a first modification of the present embodiment will be described with reference to FIG. 5. FIG. 5 is a view for explaining a configuration of a first plunger according to a first modification. Note that the same components as the components described above in FIG. 2 or the like are denoted by the same reference numerals. In the above-described embodiment, it has been described that two flat surface portions **21c** are provided in the flange portion **21b**, but in the present first modification, one flat surface portion **21c** is formed.

[0040] In a first plunger **21A** according to the first modification, one flat surface portion **21c** is provided in the flange portion **21b**, and the flat surface portion **21c** is brought into contact with the wall surface of the hole of the probe holder **3**, whereby the rotation of the probe **2a** around the axis may be controlled.

[0041] According to the above-described first modification, similarly to the above-described embodiment, in the flange portion **21b** of the columnar first plunger **21**, one planar flat surface portion **21c** is formed, and the flat surface portion **21c** abuts on the wall surface of the hole of the probe holder **3**. Therefore, by controlling the rotation of the probe **2a** around the axis while securing the volume of the distal end portion **21a**, it is possible to secure the strength of the contact probe while suppressing the rotation with respect to the probe holder.

[0042] Next, a second modification of the present embodiment will be described with reference to FIG. 6. FIG. 6 is a view for explaining a configuration of a first plunger according to a second modification. Note that the same components as the components described above in FIG. 2 or the like are denoted by the same reference numerals. In the second modification, the flange portion **21b** is further provided with a second flat surface portion **21d**.

[0043] In a first plunger **21B** according to the second modification, the flange portion **21b** includes the second flat surface portion **21d** provided at a position different from the first flat surface portion **21c** in addition to the flat surface portion **21c** (in the second modification, the first flat surface portion **21c**). The second flat surface portion **21d** is provided between the flat surface portions **21c** in the circumferential direction and has a planar shape. Note that the second flat surface portion **21d** is provided at a portion facing the other probe **2a** forming a set when installed in the probe holder **3**.

[0044] When the probe **2a** is installed in the probe holder **3**, the second flat surface portion **21d** of the probes **2a** face each other. At this time, since the rotation of the probes **2a** is regulated by the first flat surface portion **21c**, the probes **2a** may be brought close to each other to such an extent that the second flat surface portions **21d** do not come into contact with each other.

[0045] According to the above-described second modification, similarly to the above-described embodiment, in the flange portion **21b** of the columnar first plunger **21**, two planar flat surface portions **21c** are formed, and the flat surface portions **21c** abut on the wall surface of the hole of the probe holder **3**. Therefore, by controlling the rotation of the probe **2a** around the axis while securing the volume of the distal end portion **21a**, it is possible to secure the strength of the contact probe while suppressing the rotation with respect to the probe holder.

[0046] In addition, according to the second modification, since the second flat surface portions **21d** of the probes **2a** are arranged to face each other, it is possible to narrow the

pitch between the probes **2a** as compared with the configuration according to the embodiment.

[0047] Note that, in the second modification, the configuration of the first modification may be adopted, and a configuration including one first flat surface portion **21c** may be adopted.

[0048] Next, a third modification of the present embodiment will be described with reference to FIGS. 7 and 8. FIGS. 7 and 8 are perspective views for explaining a configuration of the probe according to the third modification. FIGS. 7 and 8 are perspective views of the probe as viewed from different directions. Note that the same components as the components described above in FIG. 2 or the like are denoted by the same reference numerals. In the third modification, a probe group **2A** is provided instead of the probe group **2** according to the embodiment. Note that, in FIGS. 7 and 8, only a part of the probe is illustrated.

[0049] In the probe group **2A**, two probes **2b** formed using a conductive material are arranged in parallel so as to have the same height. In addition, the probe groups **2A** are arranged according to the arrangement of the connection electrodes (leads **101**) of the semiconductor package **100**.

[0050] The probe **2b** includes a first plunger **21c** that comes into contact with the lead **101** of the semiconductor package **100** when the semiconductor integrated circuit is inspected, a second plunger **22** (see FIG. 2 or the like) that comes into contact with the electrode of the circuit board **200** including the inspection circuit, and a barrel **23** to which the first plunger **21c** is attached at one end and to which the second plunger **22** is extendably coupled at the other end. The first plunger **21c** and the second plunger **22** constituting the probe **2b**, and the barrel **23** have the same axis (here, the axis **N**).

[0051] The first plunger **21c** has a columnar distal end portion **21e** in which a flat surface inclined with respect to the axis is formed, and a flange portion **21f** extending from the proximal end side of the distal end portion **21e**, at least a part of which has a larger length in a direction orthogonal to the axis than the distal end portion **21e**, substantially coaxially. The first plunger **21c** has a stepped shape at a connection portion between the distal end portion **21e** and the flange portion **21f**. Note that the first plunger **21c** is attached to the barrel **23** by fitting, welding, or the like.

[0052] The distal end portion **21e** has an inclined surface obliquely cut with respect to the axis **N** direction at the distal end, and is in contact with the lead **101** of the semiconductor package **100** at the distal end.

[0053] In addition, the distal end portion **21e** has two flat surface portions **21g** forming a flat surface parallel to the axis **N** direction. The two flat surface portions **21g** are provided on the side portions of the inclined surface at the distal end, and are provided on the opposite sides to each other with the axis **N** interposed therebetween.

[0054] The flange portion **21f** includes two first flat surface portions **21h** obtained by cutting a part of a side surface in a planar shape, and a second flat surface portion **21i** provided at a position different from the first flat surface portion **21h**. The two first flat surface portions **21h** are provided on opposite sides to each other with respect to the axis of the flange portion **21b**. The second flat surface portion **21i** is provided between the first flat surface portions **21h** in the circumferential direction.

[0055] In the third modification, the first flat surface portion **21h** is provided on the side of the inclined surface of the distal end portion **21a**.

[0056] Note that, similarly to the flat surface portion **21c**, the first flat surface portion **21h** and the second flat surface portion **21i** may have an uneven shape at a plurality of locations in the circumferential direction to the extent that the first flat surface portion **21h** and the second flat surface portion **21i** may contact the wall surface of the probe holder **3**.

[0057] In the third modification, one flat surface portion **21g** forms a flat surface portion flush with one first flat surface portion **21h**, and the other flat surface portion **21g** forms a flat surface portion flush with the other first flat surface portion **21h**.

[0058] When the probe **2b** is installed in the probe holder **3**, the second flat surface portion **21i** of the probes **2b** face each other. At this time, since the rotation of the probes **2b** is regulated by the flat surface portion **21g** and the first flat surface portion **21h**, the probes **2b** may be brought close to each other to such an extent that the second flat surface portion **21i** do not come into contact with each other.

[0059] In the third modification, by providing the flat surface portion **21g** forming the same plane as the first flat surface portion **21h** and securing the arrangement region in the direction orthogonal to the direction in which the flat surface portion **21g** faces, it is possible to secure the volume of the distal end portion **21e** while securing the function of regulating the rotation of the probe by the first flat surface portion **21h** of the flange portion **21f**.

[0060] According to the above-described third modification, similarly to the above-described embodiment, in the flange portion **21f** of the columnar first plunger **21c**, two planar first flat surface portions **21h** are formed, and the first flat surface portions **21h** abut on the wall surface of the hole of the probe holder **3**. Therefore, by controlling the rotation of the probe **2b** around the axis while securing the volume of the distal end portion **21e**, it is possible to secure the strength of the contact probe while suppressing the rotation with respect to the probe holder.

[0061] Furthermore, according to the third modification, since the flat surface portion **21g** continuous with the first flat surface portion **21h** is formed at the distal end portion **21e** of each probe **2b**, the range having the function of regulating the rotation of the probe **2b** around the axis may be expanded, and the rotation of the probe **2b** around the axis may be regulated more reliably.

[0062] In addition, according to the third modification, since the second flat surface portion **21i** of the probes **2a** are arranged to face each other, it is possible to narrow the pitch between the probes **2b** as compared with the configuration according to the embodiment.

[0063] Although the embodiments for carrying out the present disclosure have been described so far, the present disclosure should not be limited only by the above-described embodiments. The (first) flat surface portion and the second flat surface portion are not limited to those shown in the above-described embodiment and modification, and the setting of the formation range of the flat surface portion in the flange portion and the like may be appropriately changed. In addition, the flat surface portion may be formed outside the position where the flange portion is connected to the distal end portion. That is, at the flat surface portion forming

position, a stepped shape in which the flange portion side protrudes from the distal end portion to the flange portion is formed.

[0064] In addition, the probe **2a** used for the probe group **2** is not limited to the above-described one including the plunger and the barrel as long as it has a flange portion, and may be one including the above-described plunger and coil spring having a distal end portion, or a pogo pin. In addition, the connection electrode may have a hemispherical shape in addition to a flat plate shape such as the lead **101**.

[0065] As described above, the probe unit and the contact probe according to the present disclosure are useful for ensuring the strength of the contact probe while suppressing the rotation with respect to the probe holder.

[0066] According to the present disclosure, it is possible to secure the strength of the contact probe while suppressing the rotation with respect to the probe holder.

[0067] All examples and conditional language recited herein are intended for pedagogical purposes of aiding the reader in understanding the disclosure and the concepts contributed by the inventor to further the art, and are not to be construed as limitations to such specifically recited examples and conditions, nor does the organization of such examples in the specification relate to a showing of the superiority and inferiority of the disclosure. Although the embodiments have been described in detail, it should be understood that the various changes, substitutions, and alterations could be made hereto without departing from the spirit and scope of the disclosure.

What is claimed is:

1. A probe unit comprising:

a plurality of probe groups each including two contact probes configured to come into contact with one electrode of a contact target on one end portion side in a longitudinal direction, each of the two contact probes coming into contact with a different electrode of a substrate on another end portion side; and

a probe holder configured to hold the contact probes, the probe holder including a plurality of holder holes configured to hold the plurality of contact probes, wherein each of the two contact probes includes:

a distal end portion configured to come into contact with the one electrode of the contact target at a distal end; and

a flange portion extending from a proximal end side of the distal end portion and having a maximum length in a direction orthogonal to a longitudinal axis of the contact probe larger than a maximum length of the distal end portion, the flange portion including a first flat surface portion having a planar shape provided on a part of a side surface of the flange portion, and the holder hole has a stepped shape configured to be locked by the flange portion and has a wall surface abutting on the first flat surface portion.

2. The probe unit according to claim **1**, wherein the first flat surface portion includes two flat surfaces provided on opposite sides to each other with respect to the longitudinal axis.

3. The probe unit according to claim **1**, wherein the first flat surface portion is constituted of one flat surface.

4. The probe unit according to claim **1**, wherein the flange portion includes a second flat surface portion facing another one of the two contact probes at a position different from the first flat surface portion in a state where the two contact probes are installed in the probe holder.

5. A contact probe for coming into contact with one electrode of a contact target on one end portion side in a longitudinal direction, the contact probe comprising:

a distal end portion configured to come into contact with the one electrode of the contact target at a distal end; and

a flange portion extending from a proximal end side of the distal end portion and having a maximum length in a direction orthogonal to a longitudinal axis of the contact probe larger than a maximum length of the distal end portion, the flange portion including a flat surface portion having a planar shape provided on a part of a side surface of the flange portion.

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